Day 2: TCP/IP Suite

Name: HET ACHARYA

Course: Diploma in Cyber Security

Topic: Day 2: TCP/IP Suite

Introduction

The TCP/IP Suite is a group of networking protocols used for communication on the Internet and in many local networks. It is usually explained using four layers: Application, Transport, Internet and Network Access.

The TCP/IP model is simpler than the OSI model. Protocols such as HTTP/HTTPS, TCP, UDP and IP are commonly used in these layers.

Task 1

PRACTICAL

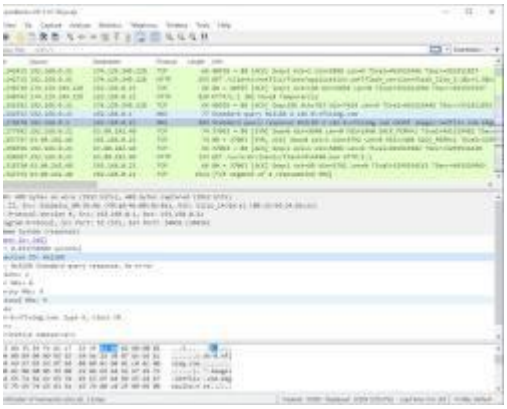
Open Wireshark and start a capture. Refresh the Flipkart homepage and stop the capture after the page loads.

Look at packet headers and identify:

• Application – HTTP/HTTPS or another application protocol visible in your capture.

• Transport – TCP or UDP.

• Internet – IP/IPv6.
• Network Access – Ethernet, Wi-Fi or another link-layer protocol shown by your capture.

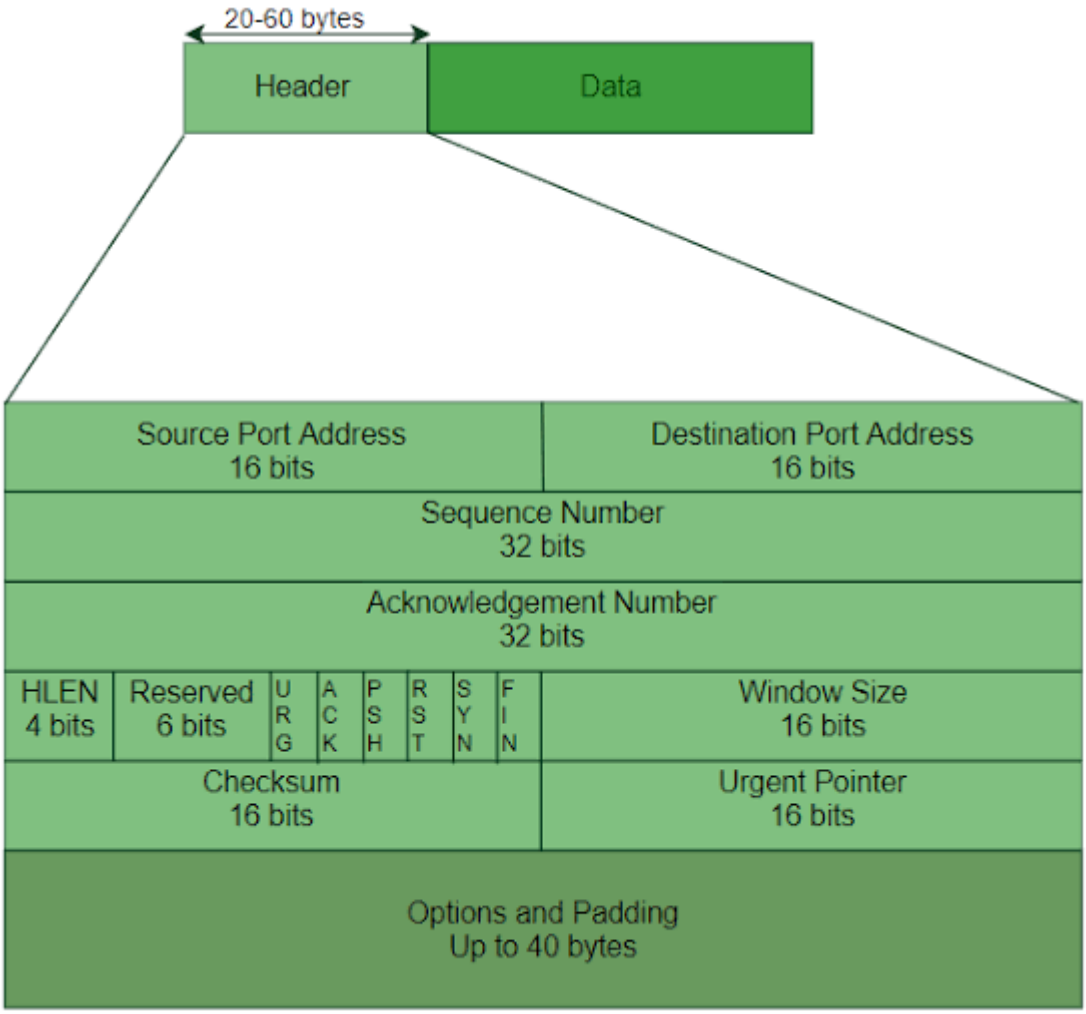


Do not write a packet as a result unless you actually see it in your capture.

Task 2

OSI Model	TCP/IP Model	Simple Example
Application	Application	WhatsApp or web browser
Presentation	Application	Data format/encryption
Session	Application	Managing a communication session
Transport	Transport	TCP/UDP
Network	Internet	IP addressing/routing
Data Link	Network Access	Ethernet/Wi-Fi frames
Physical	Network Access	Cable/radio signal

Task 3



Use the TCP segment header screenshot provided by your trainer. Label the Transport-layer fields such as Source Port, Destination Port, Sequence Number, Acknowledgment Number, Flags, Window Size and Checksum if they are visible.

Brief explanations:

• Source Port: Identifies the sending application port.

• Destination Port: Identifies the receiving application port.

• Sequence Number: Helps TCP keep data in the correct order.

• Acknowledgment Number: Shows the next data number the receiver expects.

• Flags: Control TCP actions such as starting or ending a connection.

• Window Size: Shows how much data the receiver can handle at that time.

• Checksum: Helps detect errors in the TCP segment.

Use only the fields visible in your trainer's screenshot.

Task 4

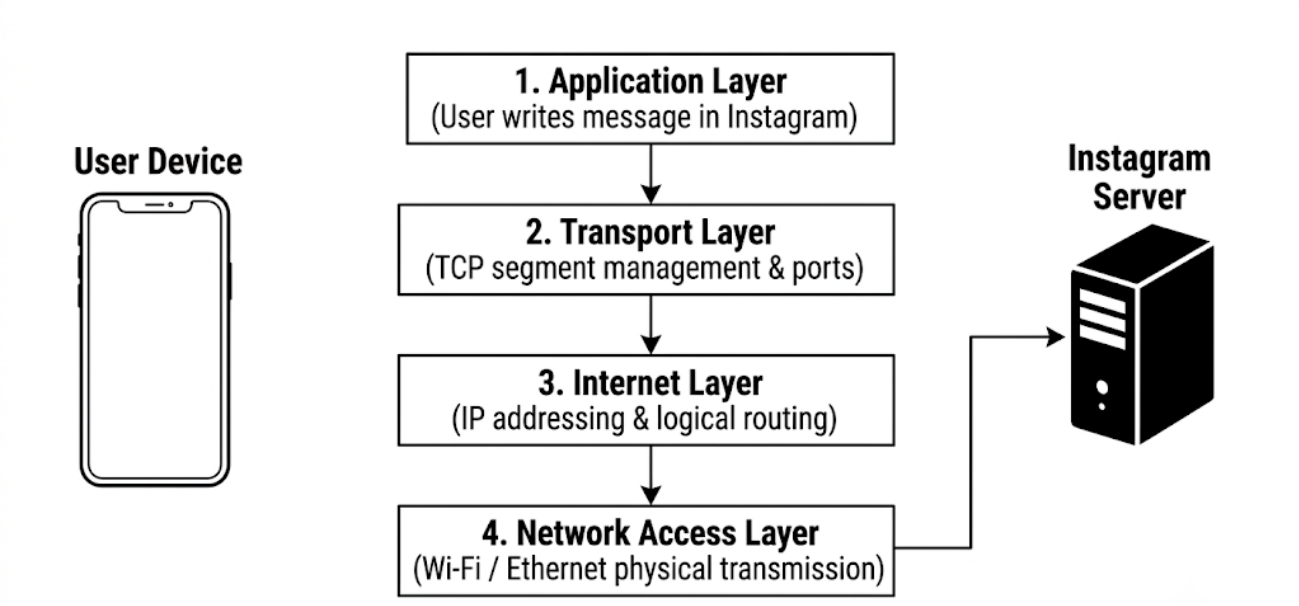


Figure: Four-Layer TCP/IP Model Instagram Communication Flow

Draw four boxes for the TCP/IP layers:
1. Application – the user writes a message in Instagram.
2. Transport – the data is managed for transport using TCP or UDP as appropriate.
3. Internet – IP addressing helps send the data toward Instagram's server.
4. Network Access – the data is placed onto Wi-Fi/Ethernet and transmitted.

Add arrows from your device to the Instagram server.

Task 5

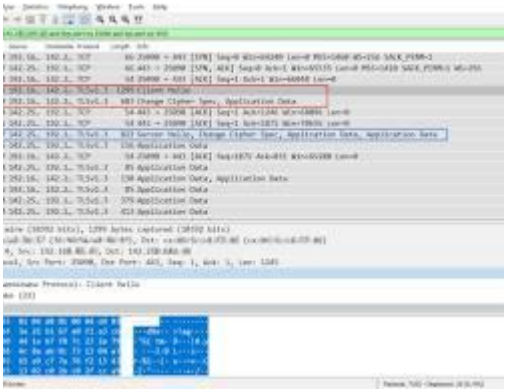
PRACTICAL

Pick one packet from your own Wireshark capture. Write:
• Packet/protocol name: TLSv1.3 (Application Data)

• TCP/IP layer: Application Layer (encapsulated over Transport/TCP)

• How I identified it: Filtered for 'tls || tcp.port == 443' and verified 'TLSv1.3' in the Protocol column with 'Application Data' in the Info column.

• Real-world action: Transmitting encrypted user feed/message data while browsing the platform securely over HTTPS.



Screenshot: Include the selected packet in Wireshark.

Do not invent the packet details.

Conclusion

The TCP/IP Suite is the main practical model behind Internet communication. Its four layers make it easier to understand how an application uses transport, IP addressing and the network connection to send data.

Quick Revision

- TCP/IP commonly uses 4 layers.
- TCP and UDP are Transport-layer protocols.
- IP is used at the Internet layer.
- The Network Access layer deals with the local connection.
- OSI has 7 layers and TCP/IP commonly has 4.